

[Linux Processes & Network Investigation Notes]

[Linux Processes & Process Management]

[What is a Process?]

A process is simply a program in execution. When you run a command or open an application, Linux creates a process for it. Each process is assigned a unique number called a **PID (Process ID)**, which the system uses to track it.

[Types of Processes]

- **Foreground Processes:** Run directly in the terminal; you must wait for them to finish before typing the next command.
- **Background Processes:** Run independently of the terminal interface, allowing you to continue working while they execute.

[Process States]

- **Running (R):** The process is either currently running or ready to run on the CPU.
- **Sleeping (S):** The process is waiting for an event or resource to become available.
- **Stopped (T):** The process has been paused or stopped by a user or a signal.
- **Zombie (Z):** The process has terminated, but its parent process hasn't properly cleaned up its exit status yet.

[Essential Process Commands]

Command	Description	Common Flags / Usage
ps	Snapshot of current processes.	<code>ps aux</code> (Shows every running process from all users).
top	Real-time interactive process monitor.	Displays CPU, memory usage, and dynamic process lists.
htop	An enhanced, user-friendly version of top.	Color-coded, supports mouse clicks, and easier to read.
kill	Sends a termination signal to a process using its PID.	<code>kill <PID></code> (Graceful stop), <code>kill -9 <PID></code> (Forceful kill).
pkill	Kills a process using its name instead of its PID.	<code>pkill firefox</code>

[Linux Networking & Investigation Commands]

System and network auditing rely heavily on observing active connections, open ports, and configurations. Niche tools help map network behavior:

[1. ifconfig / ip a]

- Used to display network interface configurations, including IP addresses, MAC addresses, and subnet masks.
- `ip a` is the modern replacement for the older `ifconfig` command.

[2. netstat]

- Displays routing tables, interface statistics, and active network connections.
- **Important Audit Flags:** `netstat -tuln`
 - `-t`: TCP connections
 - `-u`: UDP connections
 - `-l`: Listening ports
 - `-n`: Shows numeric addresses and port numbers instead of resolving names.

[3. ss]

- A faster, more modern alternative to `netstat` for dumping socket statistics.
- Commonly used as `ss -tuln` to view open sockets and listening ports quickly.

[4. lsof (List Open Files)]

- In Linux, everything is a file, including network connections. `lsof` tracks which files or network sockets are opened by which processes.
- **Audit Usage:** `lsof -i :22` checks exactly which process is interacting with Port 22 (SSH).

[Service Management (systemctl)]

Services are background processes (daemons) that start up with the system or run continuously in the background. `systemctl` is the tool used to control them:

- **Check Service Status:** `systemctl status ssh`
- **Start a Service:** `systemctl start ssh`
- **Stop a Service:** `systemctl stop ssh`
- **Restart a Service:** `systemctl restart ssh`
- **Enable on Boot:** `systemctl enable ssh`
- **Disable from Boot:** `systemctl disable ssh`

[Automated Tasks with Cron Jobs]

[What is a Cron Job?]

A cron job is a time-based job scheduler in Unix-like operating systems. It allows users to schedule tasks (commands or shell scripts) to run automatically at specific times, dates, or intervals.

[Core Commands]

- `crontab -e`: Edit the current user's cron schedules.
- `crontab -l`: List all scheduled tasks for the current user.

[Cron Syntax Format]

A cron entry uses five asterisks followed by the command to execute:

```
* * * * */path/to/command  
| | | | |  
| | | | |
```

